APPENDIX

SECTION 2.1

Figure 2.1.1 - ER MODEL

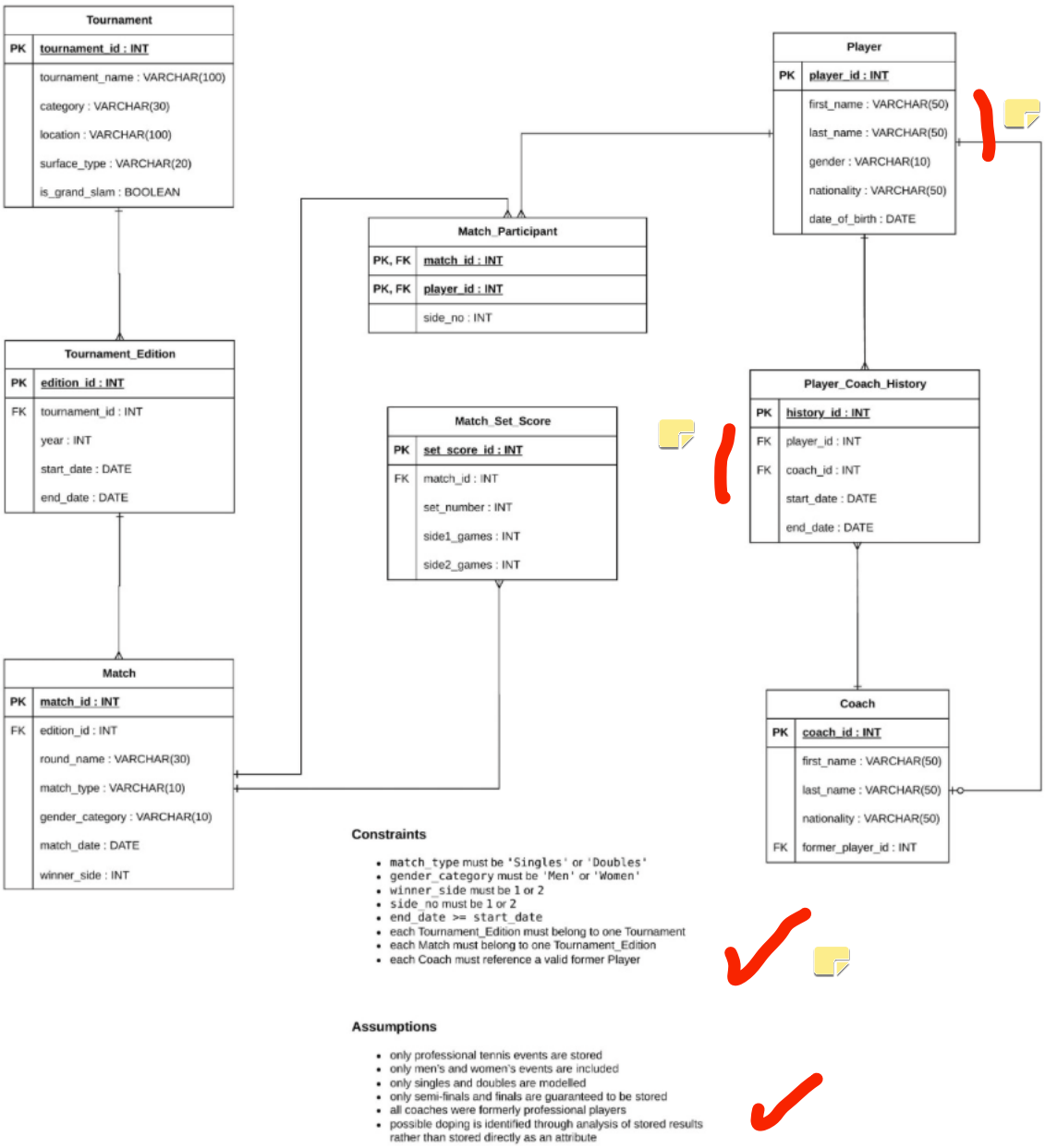


Figure 2.1.2.a: Full entity specification table showing entity purposes, attributes, primary keys, foreign keys and important constraints.

Entity	Purpose	Attributes and domains	Primary key	Foreign key(s)	Important constraints / notes
Tournament	Stores the general details of each tennis tournament.	tournament_id INT, tournament_name VARCHAR(100), category VARCHAR(30), location VARCHAR(100), surface_type VARCHAR(20), is_grand_slam BOOLEAN	tournament_id	None	Stores tournament-level details that do not change for each yearly edition, such as name, location, surface type and whether it is a Grand Slam.
Tournament_Edition	Stores a specific yearly edition of a tournament.	edition_id INT, tournament_id INT, year INT, start_date DATE, end_date DATE	edition_id	tournament_id → Tournament(tournament_id)	Each tournament edition must belong to one tournament. end_date should be greater than or equal to start_date. This separates the repeating yearly event from the general tournament information.
Match	Stores details of each tennis match in a tournament edition.	match_id INT, edition_id INT, round_name VARCHAR(30), match_type VARCHAR(10), gender_category VARCHAR(10), match_date DATE, winner_side INT	match_id	edition_id → Tournament_Edition(edition_id)	Each match must belong to one tournament edition. match_type must be either 'Singles' or 'Doubles'. gender_category must be either 'Men' or 'Women'. winner_side must be 1 or 2.
Player	Stores details of each professional tennis player.	player_id INT, first_name VARCHAR(50), last_name VARCHAR(50), gender VARCHAR(10), nationality VARCHAR(50), date_of_birth DATE	player_id	None	Player details are stored once and then reused through relationships with matches, coach history and coaches. This avoids repeating player information in match records.
Match_Participant	Links players to matches and records which side each player played on.	match_id INT, player_id INT, side_no INT	Composite primary key: match_id, player_id	match_id → Match(match_id); player_id → Player(player_id)	This resolves the many-to-many relationship between **Player** and **Match**. side_no must be 1 or 2. This structure allows both singles and doubles matches because multiple players can be assigned to each side.
Match_Set_Score	Stores the score for each set in a match.	set_score_id INT, match_id INT, set_number INT, side1_games INT, side2_games INT	set_score_id	match_id → Match(match_id)	Each set score belongs to one match. set_number identifies which set is being recorded. side1_games and side2_games store the games won by each side in that set.
Coach	Stores details of each coach.	coach_id INT, first_name VARCHAR(50), last_name VARCHAR(50), nationality VARCHAR(50), former_player_id INT	coach_id	former_player_id → Player(player_id)	Each coach can reference a valid former player record. This supports the assumption that all coaches were formerly professional players.
Player_Coach_History	Stores the history of coaching relationships between players and coaches.	history_id INT, player_id INT, coach_id INT, start_date DATE, end_date DATE	history_id	player_id → Player(player_id); coach_id → Coach(coach_id)	Records when a coach started and ended coaching a player. end_date should be greater than or equal to start_date. This allows a player to have different coaches over time.

Figure 2.2.1.a - Login to the university remote lab environment using Parallels Client.

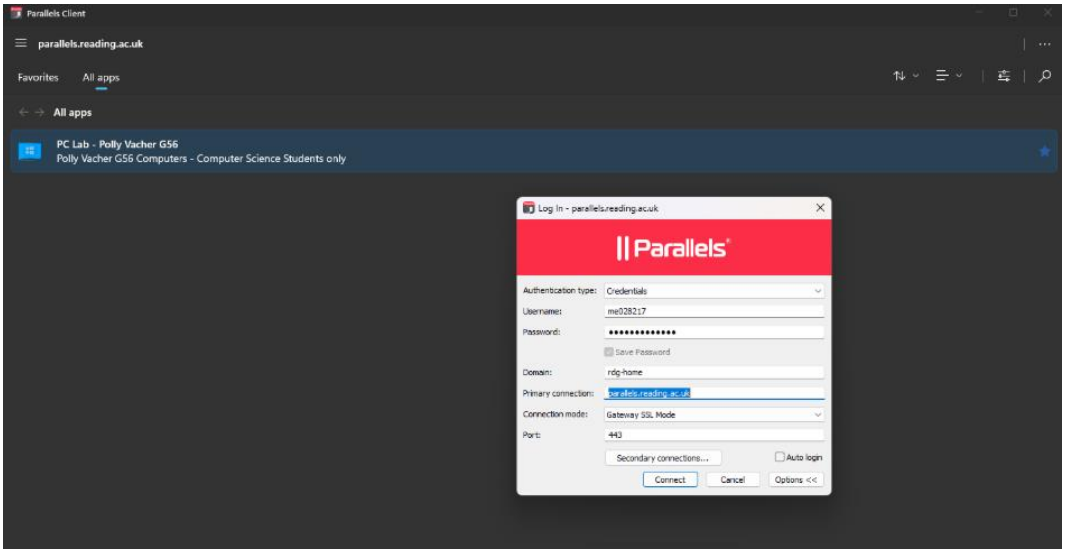
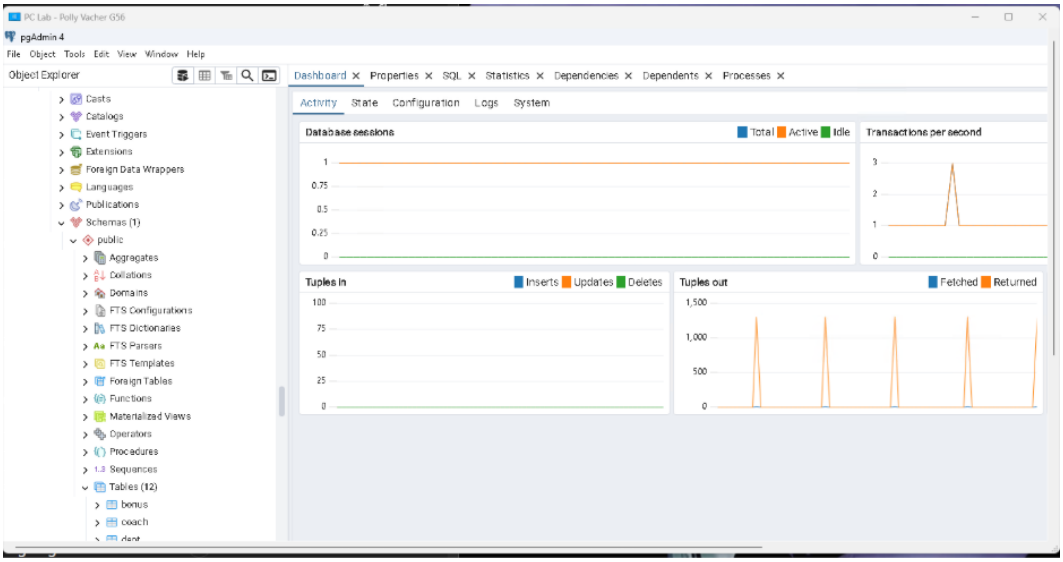


Figure 2.2.1.b - pgAdmin 4 environment used to connect to PostgreSQL and manage the tennis database.



SECTION 2.2

Figure 2.2.3.a - Player table structure/output

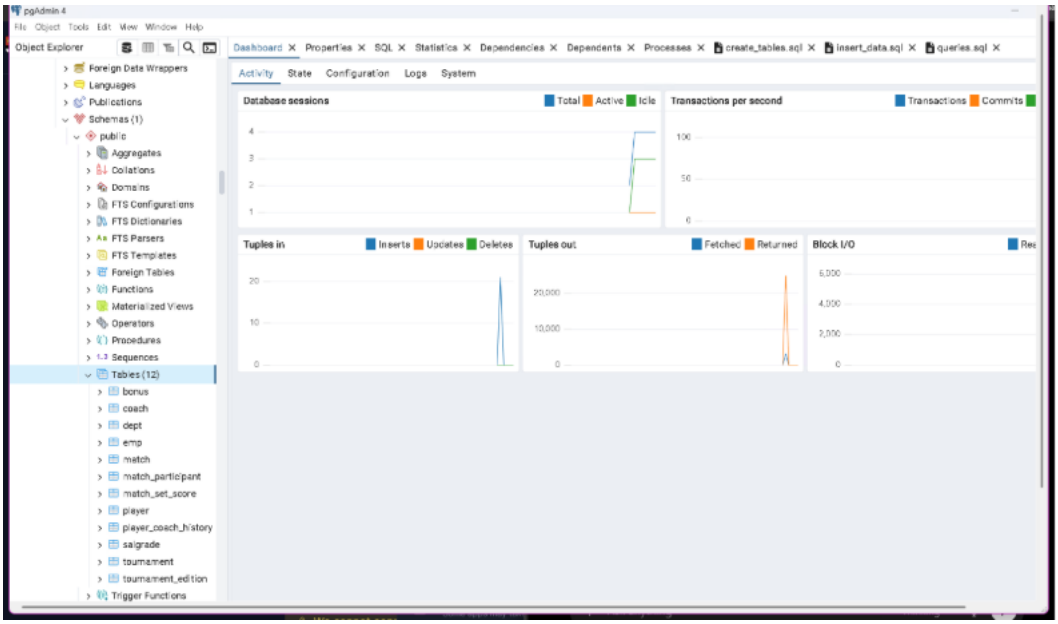


Figure 2.2.3.b - Coach table structure/output

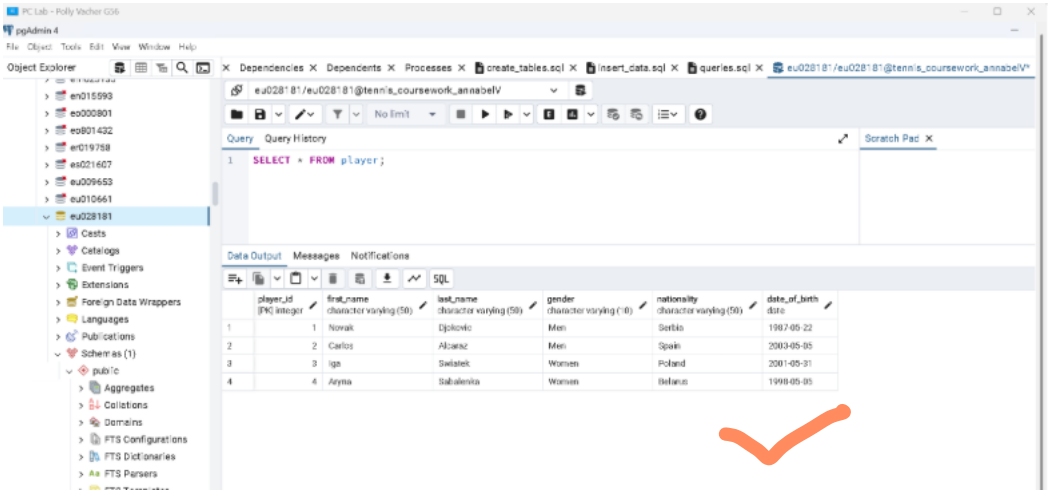


Figure 2.2.3.c - Tournament table structure/output

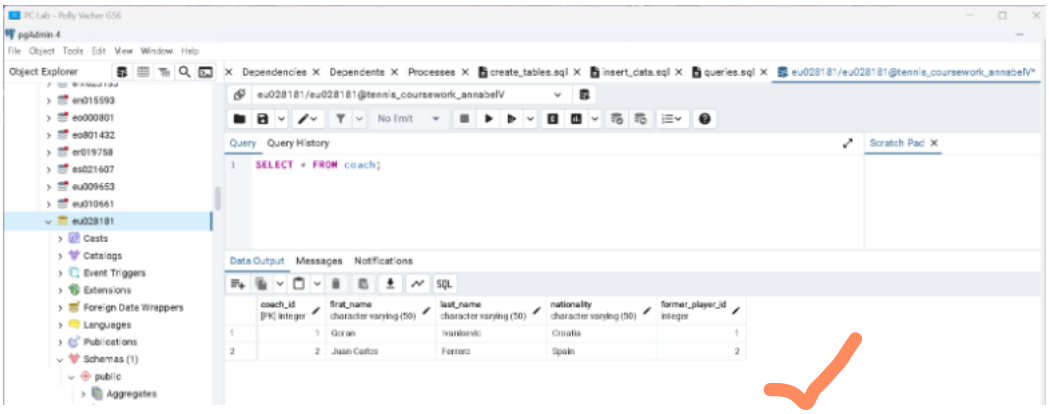


Figure 2.2.3.d - Tournament_Edition table structure/output

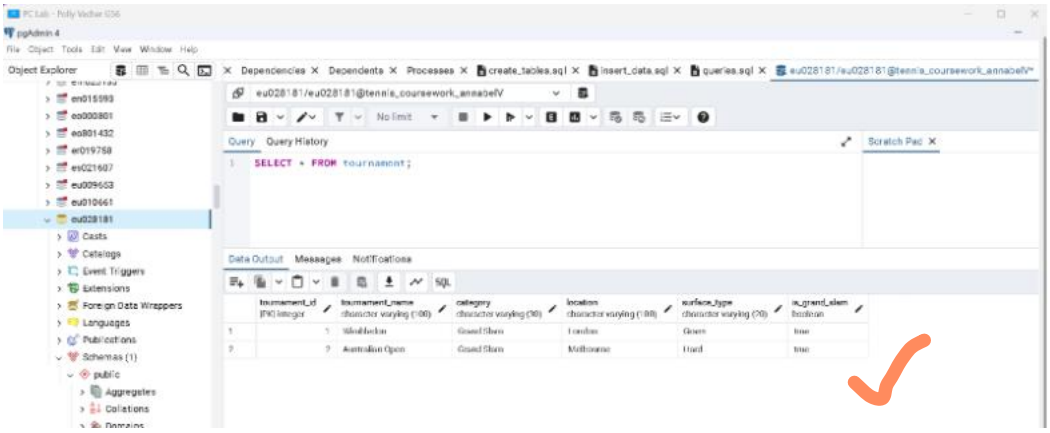


Figure 2.2.3.e - Match table structure/output

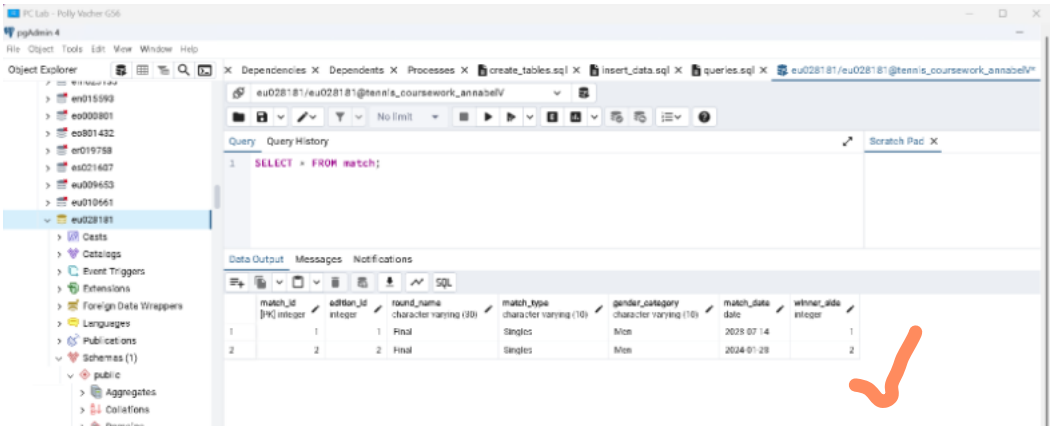


Figure 2.2.3.f - Match_Participant table structure/output

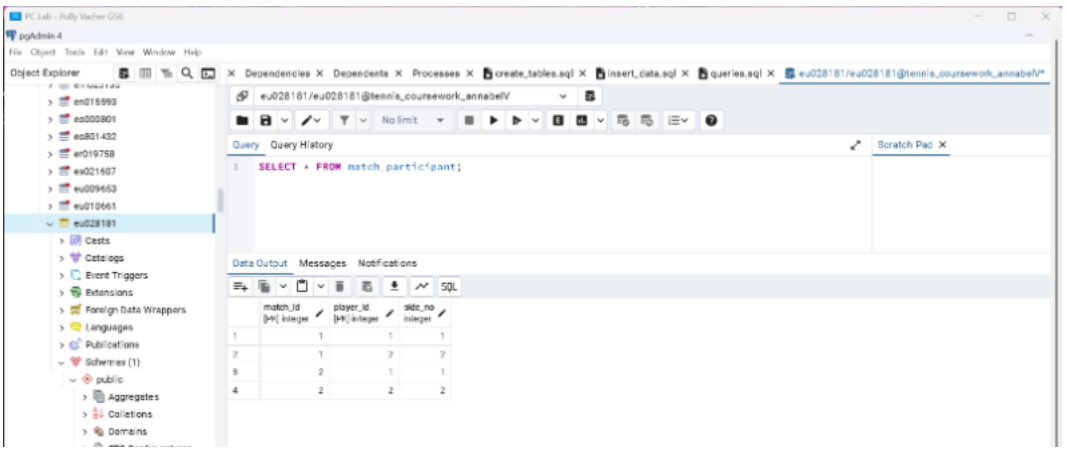


Figure 2.2.3.g - player_Coach_History / Match_Set_Score structure/output

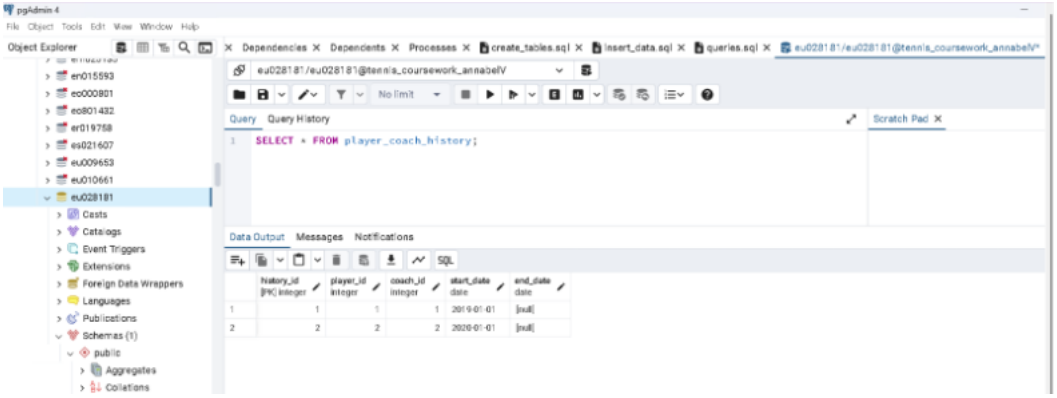


Figure 2.2.4.a - Example SQL-DML `INSERT INTO` statements used to populate the database tables.

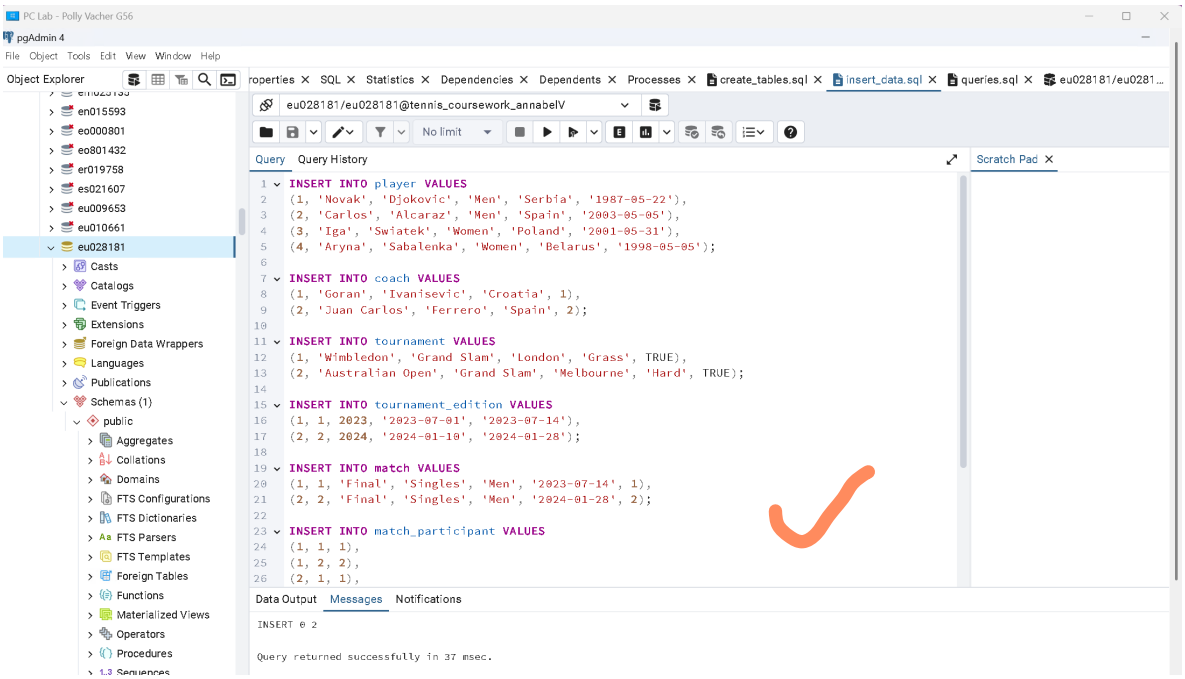


Figure 2.2.5.a - Record-count validation showing that key tables contain inserted records.

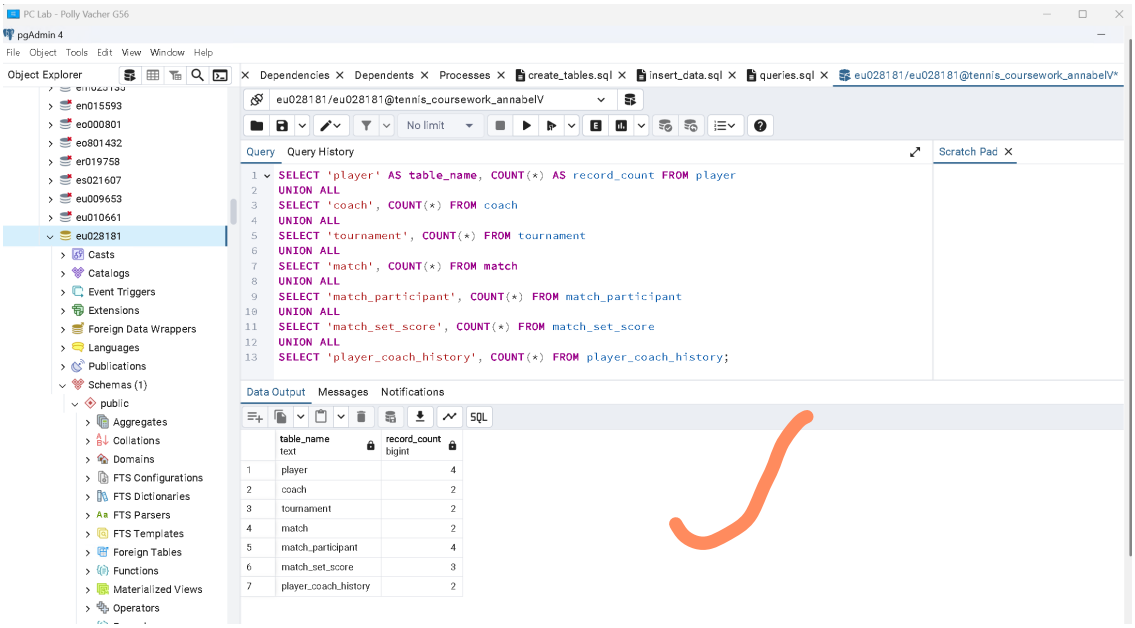


Figure 2.2.6.a - pgAdmin table list showing all 8 tables created.

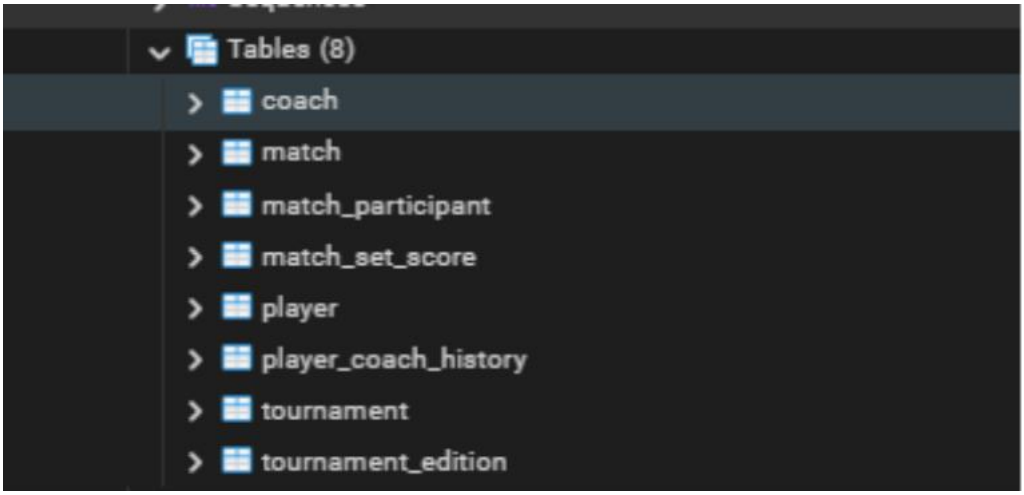


figure 2.2.6.b - CREATE TABLE SQL running successfully.

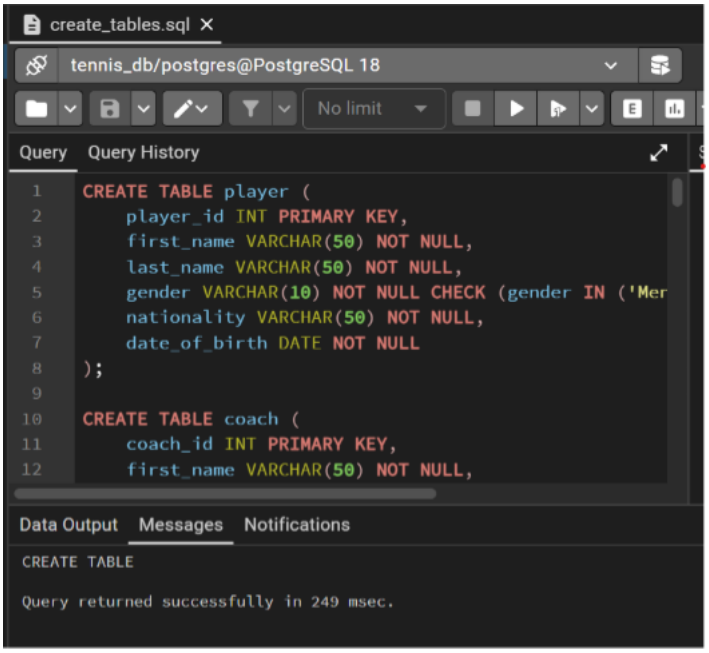


Figure 2.2.6.c - table structure/columns/keys/constraints expanded in pgAdmin.

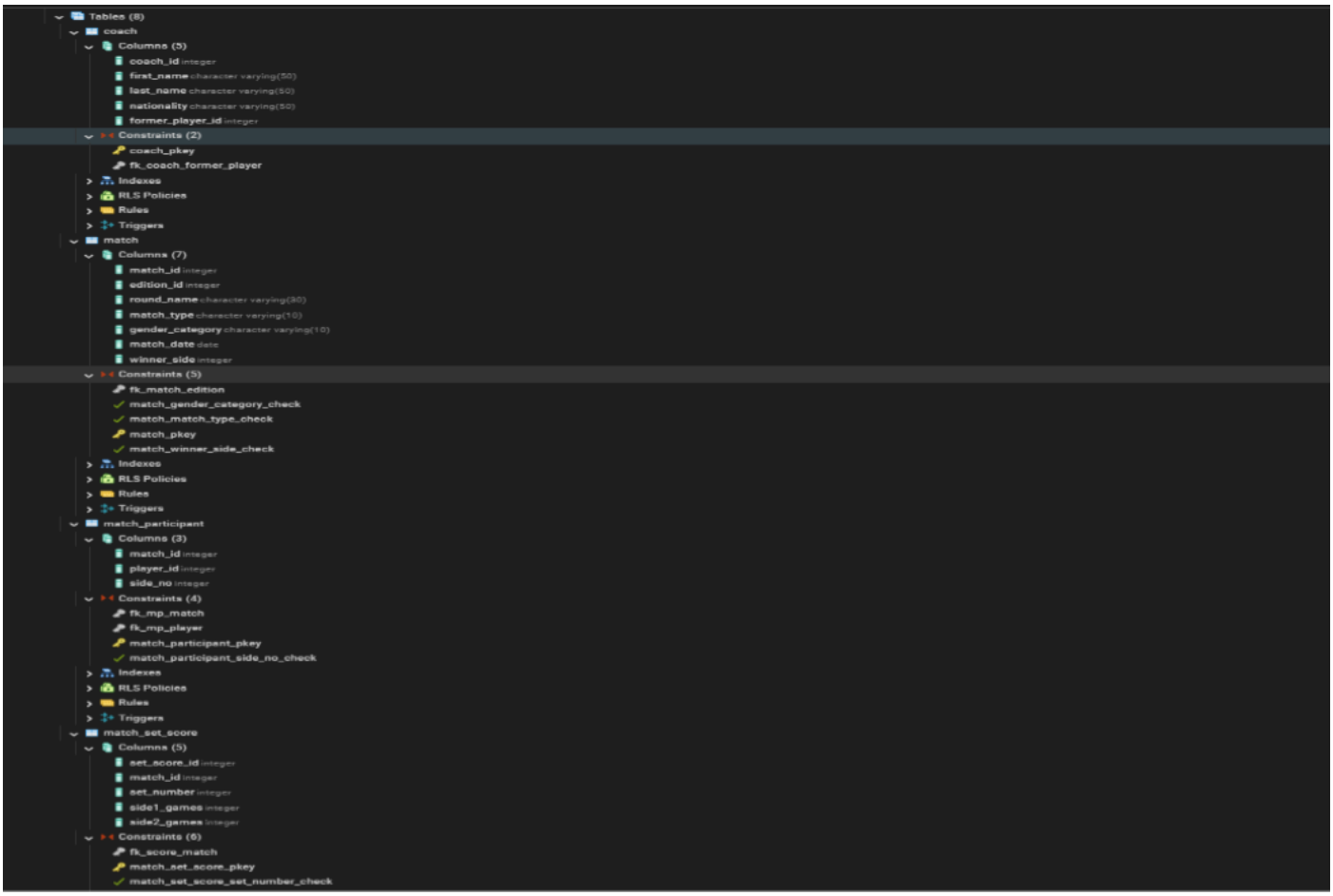


Figure 2.2.6.d - query checking match counts by tournament.

Query Query History

```
*/
-- Query 8: Count matches by tournament
SELECT
    t.tournament_name,
    COUNT(m.match_id) AS total_matches
FROM tournament t
JOIN tournament_edition te
ON t.tournament_id = te.tournament_id
JOIN match m
ON te.edition_id = m.edition_id
GROUP BY t.tournament_name;

/*
-- Query 9: Count wins by player
SELECT
    p.first_name,
    p.last_name,
```

Data Output Messages Notifications

Showing rows:

	tournament_name character varying (100)	total_matches bigint
1	Wimbledon	1
2	Australian Open	1

Figure 2.2.6.e - query checking players and their coach links.

```
-- Query 4: Show players and their coaches
SELECT
    p.first_name,
    p.last_name,
    c.first_name AS coach_first_name,
    c.last_name AS coach_last_name
FROM player p
JOIN coach c ON p.player_id = c.former_player_id;
SELECT
    p.first_name,
    p.last_name,
    c.first_name AS coach_first_name,
    c.last_name AS coach_last_name
FROM player p
JOIN coach c ON p.player_id = c.former_player_id;
```

utput Messages Notifications

Showing rows: 1 to 2

first_name character varying (50)	last_name character varying (50)	coach_first_name character varying (50)	coach_last_name character varying (50)
Novak	Djokovic	Goran	Ivanisevic
Carlos	Alcaraz	Juan Carlos	Ferrero

SECTION 2.3

Figure 2.3.1.a - Simple query using WHERE to retrieve players from a specified nationality.

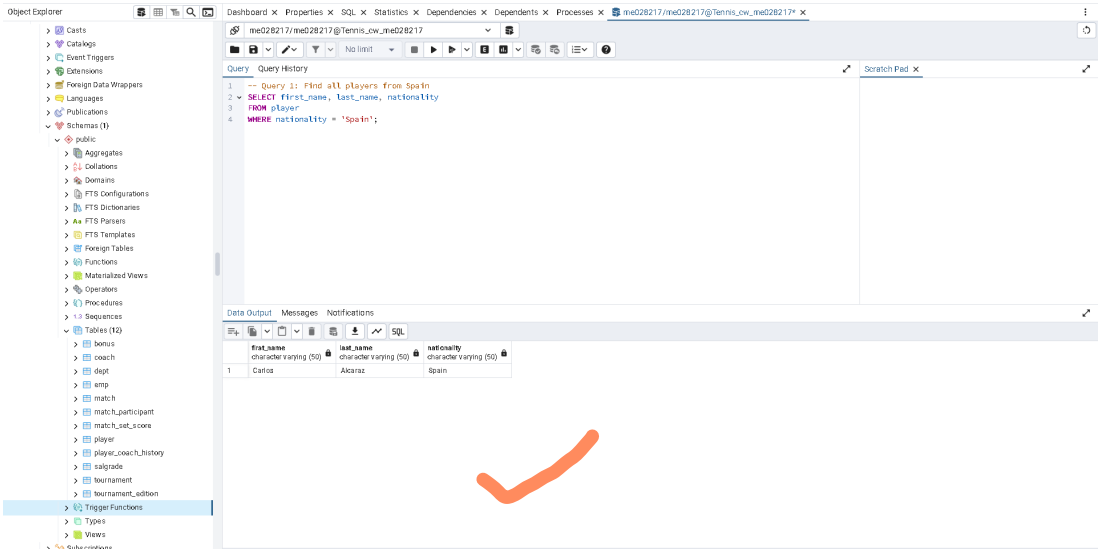


Figure 2.3.1.b - Simple grouping query using COUNT, GROUP BY, and ORDER BY to summarise players by nationality.

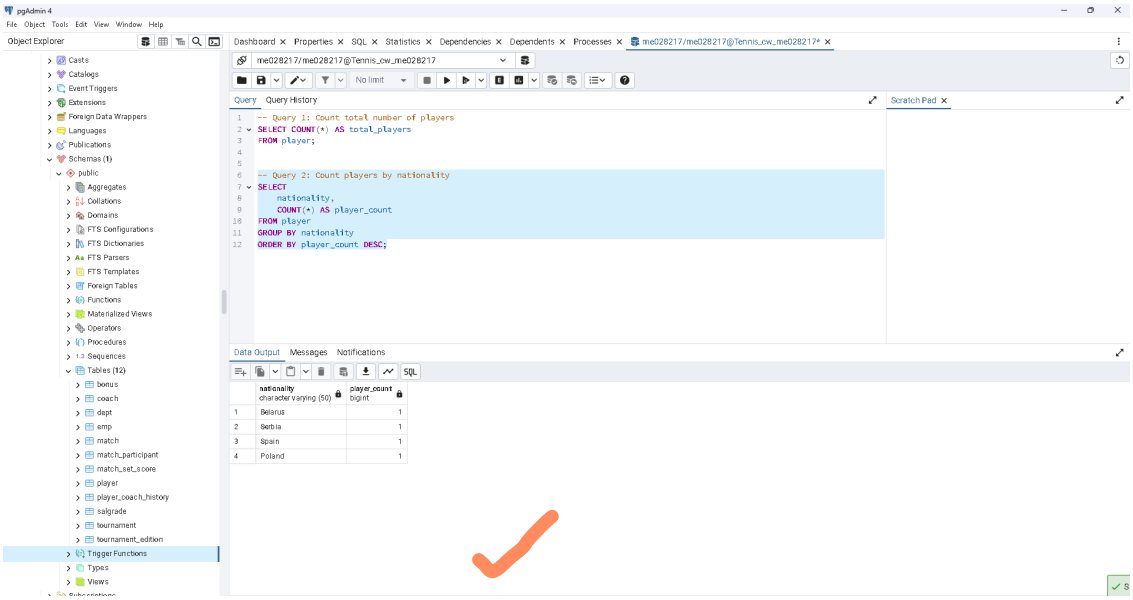


Figure 2.3.2.a - Complex query calculating match wins by player using joins between match, match_participant, and player.

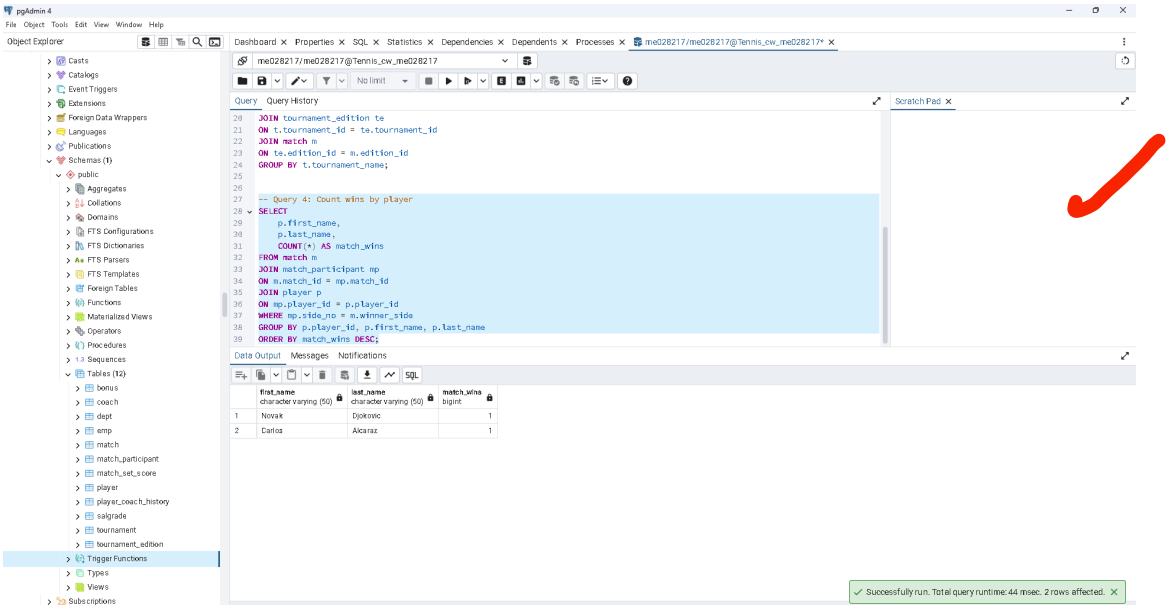
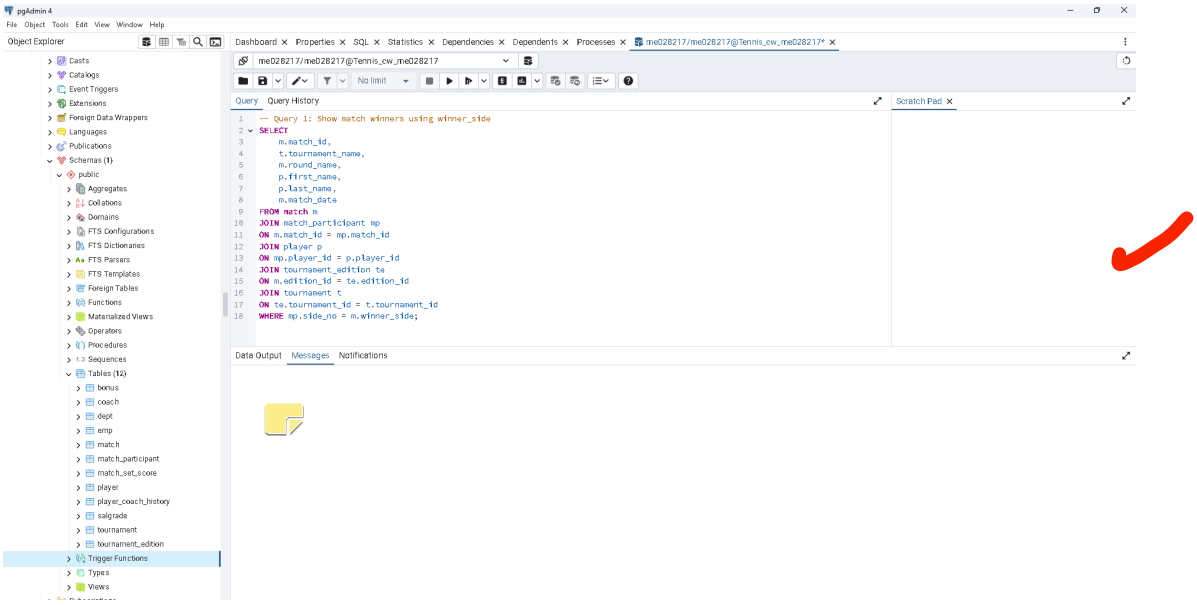


Figure 2.3.2.b - Complex multi-table query retrieving full match scorelines using match, tournament_edition, tournament, and match_set_score.



Additional Query Evidence: Further SQL query outputs used to test filtering, joins, aggregation and validation of the populated tennis database.

pgAdmin 4

me028217/me028217@Tennis_cw_me028217*

```
-- Query 1: Find all players from Spain
SELECT first_name, last_name, nationality
FROM player
WHERE nationality = 'Spain';


-- Query 2: List all female players
SELECT first_name, last_name, gender
FROM player
WHERE gender = 'Women';
```

Data Output

	first_name character varying (50)	last_name character varying (50)	gender character varying (10)
1	Iga	Swiatek	Women
2	Aryna	Sabalenka	Women

pgAdmin 4

me028217/me028217@Tennis_cw_me028217*

```
-- Query 1: Find all players from Spain
SELECT first_name, last_name, nationality
FROM player
WHERE nationality = 'Spain';


-- Query 2: List all female players
SELECT first_name, last_name, gender
FROM player
WHERE gender = 'Women';


-- Query 3: Show all tournaments played on grass courts
SELECT tournament_name, surface_type
FROM tournament
WHERE surface_type = 'Grass';
```

Data Output

	tournament_name character varying (100)	surface_type character varying (20)
1	Wimbledon	Grass

pgAdmin 4

me028217/me028217@Tennis_cw_me028217*

```
-- Query 4: Show players and their coaches
SELECT
    p.first_name,
    p.last_name,
    c.first_name AS coach_first_name,
    c.last_name AS coach_last_name
FROM player p
JOIN coach c ON p.player_id = c.former_player_id;-- Query 4: Show players and their coaches
SELECT
    p.first_name,
    p.last_name,
    c.first_name AS coach_first_name,
    c.last_name AS coach_last_name
FROM player p
JOIN coach c ON p.player_id = c.former_player_id;
```

Data Output

	first_name character varying (50)	last_name character varying (50)	coach_first_name character varying (50)	coach_last_name character varying (50)
1	Novak	Djokovic	Goran	Ivanisevic
2	Carlos	Alcaraz	Juan Carlos	Ferrero

pgAdmin 4

me028217/me028217@Tennis_cw_me028217*

```
FROM player p
JOIN coach c ON p.player_id = c.former_player_id;-- Query 4: Show players and their coaches
SELECT
    p.first_name,
    p.last_name,
    c.first_name AS coach_first_name,
    c.last_name AS coach_last_name
FROM player p
JOIN coach c ON p.player_id = c.former_player_id;


-- Query 5: Show matches with tournament names
SELECT
    m.match_id,
    t.tournament_name,
    m.round_name,
    m.match_date
FROM match m
JOIN tournament_edition te ON m.edition_id = te.edition_id
JOIN tournament t ON te.tournament_id = t.tournament_id;
```

Data Output

	match_id integer	tournament_name character varying (100)	round_name character varying (30)	match_date date
1	1	Wimbledon	Final	2023-07-14
2	2	Australian Open	Final	2024-01-28

pgAdmin 4

me028217/me028217@Tennis_cw_me028217

```
-- Query 1: Count total number of players
SELECT COUNT(*) AS total_players
FROM player;
```

	total_players bigint
1	4

pgAdmin 4

me028217/me028217@Tennis_cw_me028217

```
-- Query 2: Count players by nationality
SELECT
    nationality,
    COUNT(*) AS player_count
FROM player
GROUP BY nationality
ORDER BY player_count DESC;


-- Query 3: Count matches by tournament
SELECT
    t.tournament_name,
    COUNT(m.match_id) AS total_matches
FROM tournament t
JOIN tournament_edition te
ON t.tournament_id = te.tournament_id
JOIN match m
ON te.edition_id = m.edition_id
GROUP BY t.tournament_name;
```

	tournament_name character varying (100)	total_matches bigint
1	Wimbledon	1
2	Australian Open	1

pgAdmin 4

me028217/me028217@Tennis_cw_me028217

```
WHERE mp.side_no = m.winner_side;


-- Query 2: Show full match scoreline by set
SELECT
    m.match_id,
    t.tournament_name,
    m.round_name,
    m.match_date,
    mss.set_number,
    mss.side1_games,
    mss.side2_games
FROM match m
JOIN tournament_edition te
ON m.edition_id = te.edition_id
JOIN tournament t
ON te.tournament_id = t.tournament_id
JOIN match_set_score mss
ON m.match_id = mss.match_id
ORDER BY m.match_id, mss.set_number;
```

	match_id integer	tournament_name character varying (100)	round_name character varying (30)	match_date date	set_number integer	side1_games integer	side2_games integer
1	1	Wimbledon	Final	2023-07-14	1	6	4
2	1	Wimbledon	Final	2023-07-14	2	5	7
3	1	Wimbledon	Final	2023-07-14	3	6	3

pgAdmin 4

me028217/me028217@Tennis_cw_me028217

```
    mss.side2_games
FROM match m
JOIN tournament_edition te
ON m.edition_id = te.edition_id
JOIN tournament t
ON te.tournament_id = t.tournament_id
JOIN match_set_score mss
ON m.match_id = mss.match_id
ORDER BY m.match_id, mss.set_number;


-- Query 3: Find players who have both played matches and have coaching experience
SELECT DISTINCT
    p.first_name,
    p.last_name
FROM player p
JOIN match_participant mp
ON p.player_id = mp.player_id
JOIN coach c
ON p.player_id = c.former_player_id;
```

	first_name character varying (50)	last_name character varying (50)
1	Carlos	Alcaraz
2	Novak	Djokovic

Figure 2.3.3.a - Simple query test retrieving all female players from the player table.

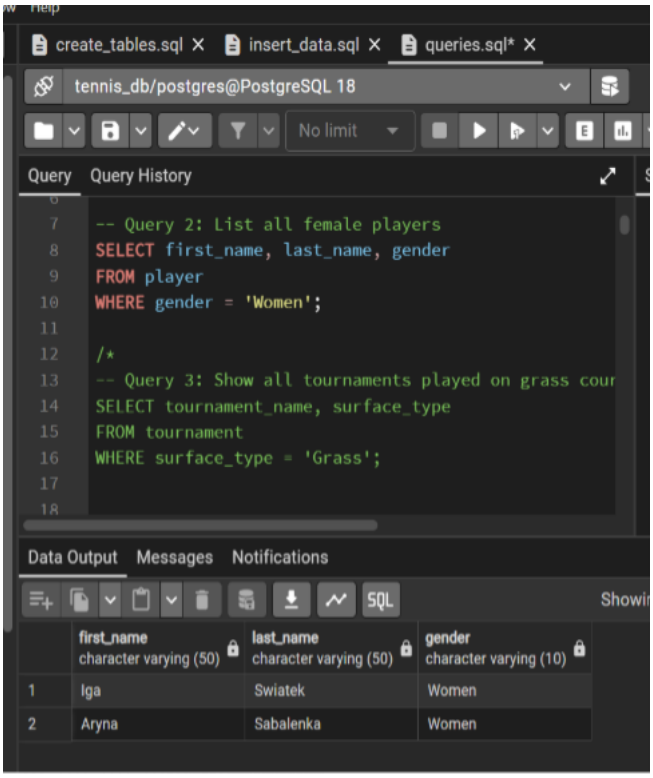


Figure 2.3.3.b - Simple query test using COUNT to validate the total number of player records.

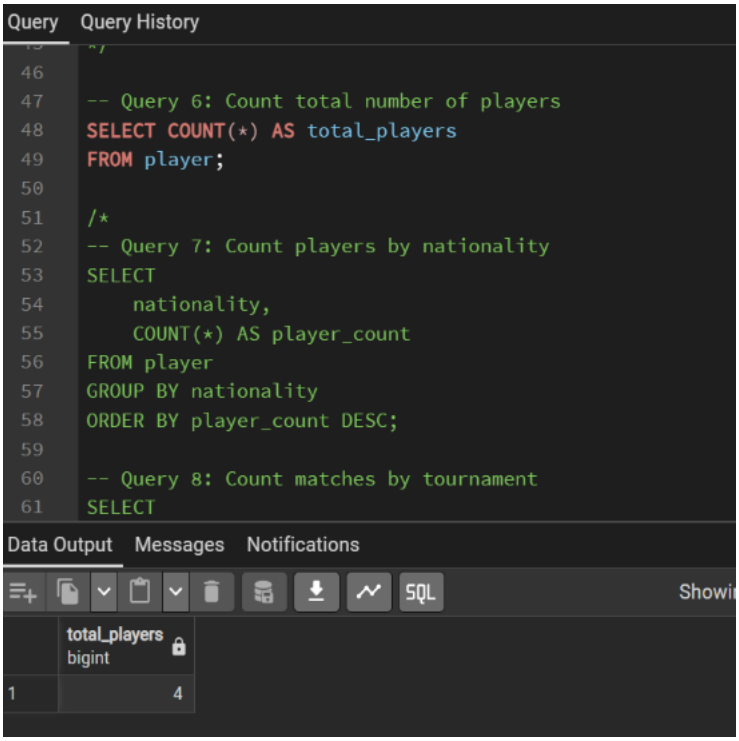


Figure 2.3.3.c - complex query test retrieving full match scorelines using joins between match, tournament_edition, tournament and match_set_score.

Query Query History

```sql
-- Query 10: Show full match scoreline by set
SELECT
    m.match_id,
    t.tournament_name,
    m.round_name,
    m.match_date,
    mss.set_number,
    mss.side1_games,
    mss.side2_games
FROM match m
JOIN tournament_edition te
ON m.edition_id = te.edition_id
JOIN tournament t
ON te.tournament_id = t.tournament_id
JOIN match_set_score mss
ON m.match_id = mss.match_id
ORDER BY m.match_id, mss.set_number;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1

	match_id integer	tournament_name character varying (100)	round_name character varying (30)	match_date date	set_number integer	side1_games integer	side2_games integer
1	1	Wimbledon	Final	2023-07-14	1	6	4
2	1	Wimbledon	Final	2023-07-14	2	5	7
3	1	Wimbledon	Final	2023-07-14	3	6	3

Figure 2.3.3.d - Complex query test calculating match wins by player using joins between match, match_participant and player.

No limit

Query Query History

```sql
*/
-- Query 9: Count wins by player
SELECT
    p.first_name,
    p.last_name,
    COUNT(*) AS match_wins
FROM match m
JOIN match_participant mp
ON m.match_id = mp.match_id
JOIN player p
ON mp.player_id = p.player_id
WHERE mp.side_no = m.winner_side
GROUP BY p.player_id, p.first_name, p.last_name
ORDER BY match_wins DESC;

/*
-- Query 10: Show full match scoreline by set
```

Data Output Messages Notifications

Showing r

	first_name character varying (50)	last_name character varying (50)	match_wins bigint
1	Novak	Djokovic	1
2	Carlos	Alcaraz	1